

Book I. Prop. XLVI. Prob.

Upon a given straight line (—) to
construct a square.



Draw — $\perp$ and = — (pr. I.11, I.3)

Draw — $\parallel$ —,
and meeting — drawn $\parallel$ —.



In

— = — (const.)

— = a right angle (const.)

$\therefore$ — = — = a right angle (pr. I.29),
and the remaining sides and angles must be equal (pr. I.34).

And $\therefore$ — is a square (def. 30).

Q. E. D.